

## AQUAFABA &amp; EGG REPLACERS

*What it is, how it works, what to use when, and why it actually foams.*

## WHAT IS IT

The liquid from cooking chickpeas. Proteins, starches, and saponins leached during cooking give it the ability to **foam, emulsify, and bind** — mimicking several structural roles of eggs. Named in 2015 (*aqua* = water, *faba* = bean). About 3–5 kcal per 2 tbsp. It's here for structure, not nutrition.

## WHY IT WORKS — THE SCIENCE

**Proteins** Legumin, vicilin, convicilin unfold under agitation and form an elastic network around air bubbles — same mechanism as egg white foam, with plant proteins.

**Starches** Increase viscosity of the liquid film around each bubble, slowing drainage and collapse.

**Saponins** Amphiphilic surfactants — hydrophobic core + hydrophilic sugar chains — that sit at the air–water interface and lower surface tension, making bubbles cheap to form.

**Cream of tartar** Lowers pH toward the proteins' isoelectric point → they aggregate more readily, form tighter foam films, resist weeping (syneresis). Add after foam forms, before soft peaks. ¼ tsp per egg white.

## WHIPPING GUIDE

1. Clean, completely dry metal or glass bowl. Any fat kills the foam.
2. Start on medium speed — builds small, stable bubbles first.
3. Add cream of tartar once foam forms, before soft peaks.
4. Increase to high. Expect 5–10 min to stiff peaks (vs. 3–4 for egg whites).
5. Use immediately — weeps and deflates faster than egg whites.
6. Don't overwhip. Grainy texture = gone too far. No recovery.

## IDEAL WHIPPING TEMP

70–75°F (21–24°C)

*Cold works but takes longer. Set bowl in warm water to bring up to temp.*

## NUTRITIONAL COMPARISON (PER EGG EQUIVALENT)

- **Whole egg** 6g protein · 5g fat · ~70 kcal · vitamins D, B12, choline
- **Egg white** 3.6g protein · 0g fat · ~17 kcal · complete protein
- **Aquafaba (2 tbsp)** ~0.5–1g protein · ~3–5 kcal · nutritionally negligible
- **Flax egg** 2g protein · 3g fat (omega-3s) · ~37 kcal · fiber
- **Chia egg** 2g protein · 3g fat · ~40 kcal · calcium, magnesium
- **Silken tofu (¼ cup)** 4–5g protein · 2g fat · ~35 kcal · iron, calcium
- **Banana / applesauce** ~0.5g protein · ~55 kcal · mostly sugars
- **Commercial powder** 0g protein · ~15–20 kcal · nutritionally inert

## CANNED VS. FROM SCRATCH

**Canned** — consistent, convenient, ready to use. Minimal-ingredient, no-salt-added brands perform best. If thin, reduce gently and cool.

**From scratch** — soak overnight → simmer 1.5–2 hrs → cool in liquid → strain. More concentrated, more stable foam. Result should coat a spoon lightly and feel slightly slippery.

## WHAT EGGS DO — PICK YOUR SUBSTITUTE BY FUNCTION

- **Foam** Aquafaba — the only substitute that whips. Nothing else comes close.
- **Emulsify** Aquafaba (unwhipped), silken tofu
- **Bind** Flax egg, chia egg, silken tofu, banana, applesauce
- **Leaven / Lift** Whipped aquafaba + vinegar + baking soda — combine for best results
- **Moisture** Applesauce, banana, pumpkin, yogurt
- **Richness / Fat** Silken tofu, full-fat coconut milk
- **Protein set** JUST Egg or silken tofu — quiche, frittata. Aquafaba won't help here.
- **Neutral flavor** Aquafaba, white chia, commercial replacer — avoid banana/flax in pale batters

## SUBSTITUTES — AMOUNTS &amp; BEST FOR

| SUBSTITUTE          | AMOUNT                       | BEST FOR                              |
|---------------------|------------------------------|---------------------------------------|
| ★ Aquafaba          | 2T / egg white · 3T / whole  | Meringue, mousse, dacquoise, mayo     |
| Flax egg            | 1T flax + 3T water, 5–10 min | Dense cookies, brownies, muffins      |
| Chia egg            | 1T chia + 3T water, 5–10 min | Same as flax; white chia = less color |
| Banana / applesauce | ¼ cup per egg                | Banana bread, spiced muffins          |
| Silken tofu         | ¼ cup blended per egg        | Cheesecakes, custards, quiches        |
| Vinegar + soda      | 1 tsp soda + 1 tbsp vinegar  | Lift only — pair with a binder        |
| Commercial replacer | Per package                  | Cakes and cookies                     |
| JUST Egg            | 3 tbsp per egg               | Savory: scrambles, frittatas          |

## COMMON FAILURES — CAUSE &amp; FIX

**Fat in bowl** Lipids outcompete proteins at the air–water interface → foam collapses instantly. Wipe bowl with vinegar. Metal or glass only — plastic retains fat.

**Overwhipping** Protein network over-extends → grainy, wet mass. Watch for stiffness, not time. Narrower margin than egg whites.

**Humidity** Water vapor disrupts protein concentration → foam weeps fast. Work in a cool, dry space and use immediately.

**Thin liquid** Low protein density = unstable foam. Reduce on the stovetop until slightly syrupy; cool completely before whipping.

## COMBINING SUBSTITUTES — CHEAT SHEET

**Meringue / dacquoise** Aquafaba + cream of tartar

**Light vegan cake** Aquafaba (foam) + flax egg (bind) + vinegar (lift)

**Cheesecake / custard** Silken tofu blended smooth

**Muffin / quick bread** Flax or chia egg + applesauce for moisture

**Mayo / emulsion** Unwhipped aquafaba — rivals egg yolk

**Quiche / frittata** JUST Egg — nothing else sets properly

**Bottom line:** Aquafaba is the only substitute that whips — irreplaceable for meringue-based work. Flax egg handles binding for most everyday baking. Combine substitutes for complex jobs.